Question 6

(a). Find the tensile force required to close the 4 mm gap for the system as shown in Figure 6a.

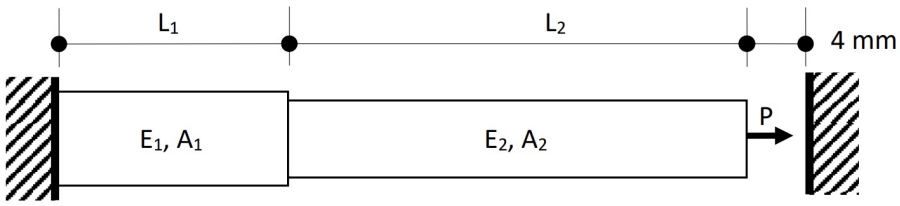


Figure 6a

The material and sectional properties of the two bars are as follows: E_1 = 120,000 MPa, E_2 = 150,000 MPa, L_1 = 2 m, L_2 = 5 m, A_1 = 300 mm^2 and A_2 = 200 mm^2.

$$E = \frac{Fx}{A\Delta x}$$
$$\Delta x = \frac{Fx}{AE}$$
$$\Delta x_1 + \Delta x_2 = 4mm$$
$$\frac{P \times 2}{0.0003 \times 120000 \times 10^6} + \frac{P \times 5}{0.0002 \times 150000 \times 10^6} = 0.004$$
$$P = 18kN$$

(b). A concrete column rested on ground is shown in Figure 6b. Three point loads are applied on the top of the column. The width of the block is 2.5 m. The concrete density is ρ = 2450 kg/m^3. Determine the stresses at A and B. (g = 10 m/s^2).

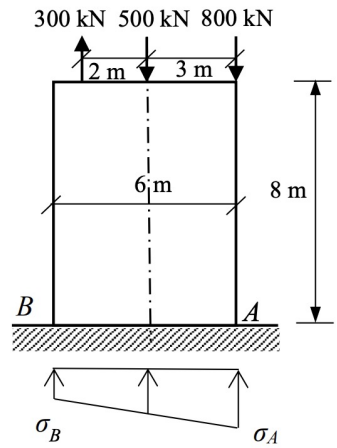


Figure 6b

Consider units in kN:

$$\sigma_{axial} = \frac{(500 + 800 - 300) + \frac{2450 \times 6 \times 8 \times 2.5 \times 10}{1000}}{6 \times 2.5} = 262.667$$

$$\sigma_{flex} = \frac{6 \times 800 \times 3 + 6 \times 300 \times 2}{2.5 \times 6^2}$$

$$= 200\ Pa$$

$$\sigma_B = \sigma_{axial} - 200 = 62.67\ Pa$$

$$\sigma_A = \sigma_{axial} + 200 = 462.67\ Pa$$

Question 7

Q7

Figure 7 shows a gate holding 4 m depth of water behind it and the inclination angle of the gate is 30° to the horizontal. The width of the gate is 5 m. The density of water is ρ = 1000 kg/m^3. The weight of the gate can be ignored. Determine the reactions at A and B. (g = 10 m/s^2)

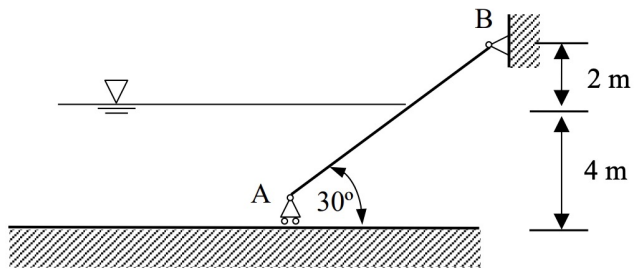


Figure 7

The force exerted onto the gate by the water pressure is:

$$\begin{aligned} F_p &= \frac{\rho g d w L}{2} \\ &= \frac{1000 \cdot 10 \cdot 4 \cdot 5 \cdot \frac{4}{\sin 30^\circ}}{2} \\ &= 800\ kN \end{aligned}$$

F_R is located at $\frac{8}{3}$ from the base.

$$L = 12m$$

$$\begin{aligned} \sum M_B &= 0 \\ F_p \times (\frac{8 \times 2}{3} + 4) &= F_A \sin(60^\circ) \times 12 \\ F_A &= 718.48\ kN \text{ up} \end{aligned}$$

$$\begin{aligned} \sum F_x &= 0 \\ F_{Bx} &= F_p \times \cos(60^\circ) \\ &= 400\ kN \text{ to the left} \\ \sum F_y &= 0 \\ F_{By} &= F_p \times \sin(60^\circ) - F_A \\ &= 25.66\ kN \text{ downward} \end{aligned}$$

Question 8

Q8

(a). Find all the stable structure(s) as shown in Figure 8a.

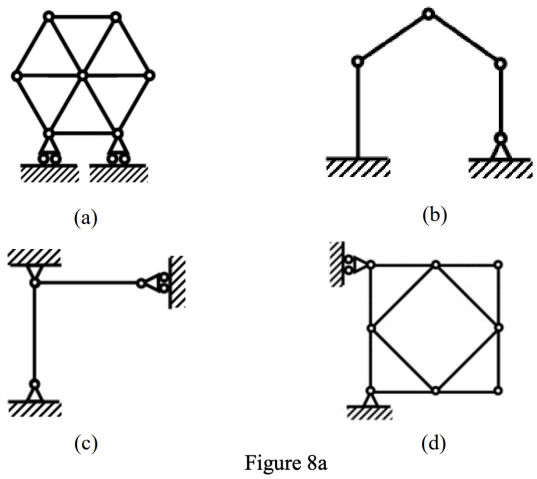


Figure 8a

There are no stable structures.

(b). Determine all the zero force members in Figure 8b.

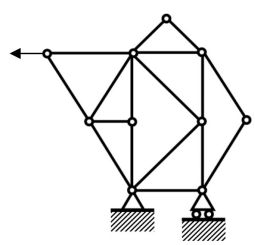
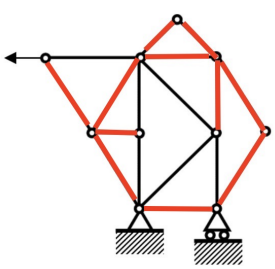
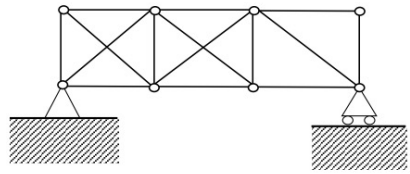


Figure 8b

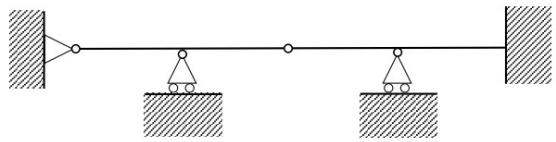
Highlighted all zero-force members.



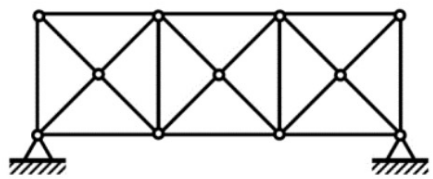
(c). Find the degree of indeterminacy of the structures in Figure 8c.



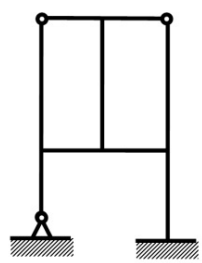
(i)



(ii)



(iii)



(iv)

i: Reaction (r) = 3, joints (j) = 8, members (m) = 15, $(r + m) > 2j \rightarrow 18 > 16$, Indeterminancy is 2
ii: r = 7, j = 4, m = 4, $(r + m) > 2j \rightarrow 11 > 8$, Indeterminancy is 3
iii: r = 4, j = 11, m = 22, $(r + m) > 2j \rightarrow 26 > 22$, Indeterminancy is 4
iv: Indeterminancy is 3+3+2 = 8

Question 9

Q9

Find the sum of internal forces of Members AB, BC and BD of the truss shown below. Please note that compression is positive.

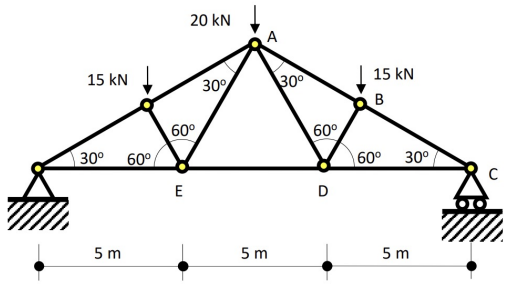


Figure 9

$$R_C = -\frac{15+20+15}{2} = -25kN$$

$$\sum F_{Cy} = 0$$
$$R_C + F_{BC}\sin 30^\circ = 0$$
$$F_{BC} = 50\ kN$$

$$\sum F_{Dy} = 0 \rightarrow (F_{BD} + F_{AD})\sin 60^\circ = 0$$
$$F_{AD} = -F_{BD}$$

$$\sum F_{Bx} = 0 \rightarrow F_{BD}\cos 60^\circ + F_{BA}\cos 30^\circ = F_{BC}\cos 30^\circ$$
$$\frac{F_{BD}}{2} + \frac{F_{BA}\sqrt{3}}{2} = \frac{50\sqrt{3}}{2}$$
$$F_{BD} = 50\sqrt{3} - F_{BA}\sqrt{3}$$

$$\sum F_{Ay} = 0 \rightarrow 2F_{AD}\sin 60^\circ + 2F_{BA}\sin 30^\circ = 20$$
$$-20 + F_{AD}\sqrt{3} + F_{BA} = 0$$
$$-20 - F_{BD}\sqrt{3} + F_{BA} = 0$$
$$-20 - (50\sqrt{3} - F_{BA}\sqrt{3})\sqrt{3} + F_{BA} = 0$$
$$-20 - 150 + 3F_{BA} + F_{BA} = 0$$
$$F_{BA} = 42.5\ kN$$

$$F_{BD} = 50\sqrt{3} - 42.5\sqrt{3}$$
$$F_{BD} = 12.99\ kN$$

Question 10

Q10

Determine the reaction force at A (Note: upward reaction is positive).

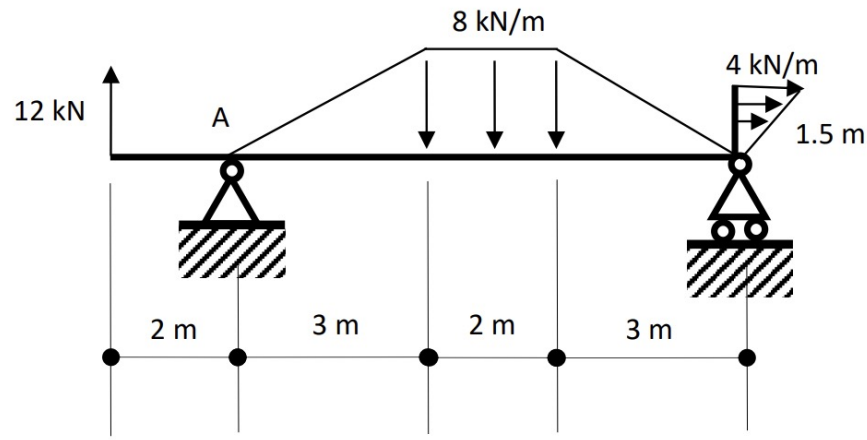


Figure 10

Consider units in kN, moment about roller:

$$F_{Ax} = \frac{4 \times 1.5}{2} = 3\ kN$$

$$(\frac{2}{3} \cdot 1.5 \times \frac{1}{2} \cdot 4 \cdot 1.5) + (8 \cdot F_{Ay}) + (12 \cdot 10) = (\frac{8 \cdot 3}{2} \times (6+2) + 8 \cdot 2 \times 4)$$

$$F_{Ay} = 4.625\ kN$$

$$F_A = \sqrt{3^2 + 4.625^2} = 5.513\ kN$$